Name: Jose Salinas
Course: CS-6370
Professor: Meilin Liu
Date: 10/13/2026
Project Name: Project 2 Tiling Matrix

Report:

Time	128*128		1024*1024		4096*4096		
CPU computation time	2.76 ms		2970.52 ms		267034.31300 ms		
GPU computation time	Tile width: 8	Tile width: 16	Tile width: 8	Tile width: 16	Tile width: 8	Tile width: 16	Tile width: 32
GPU memory transfer time	.087200 ms	.086016 ms	1.489312 ms	1.38784 ms	35.4927 ms	28.9184 ms	29.593438 ms
Speed Ups	31.65	45.877	1994.56	2140.39	7523.64	9234.063	9023.44

The tiling matrix addition program compile. All project files have course number, professor name, data, and the project name. The README will instruct how to compile and use the tiling matrix addition program. Executing the program will execute respective calculations on CPU and GPU and print the results on the screen. The tiling matrix addition program results are correct. For small matrices (8 x 8) both will print the whole matrix. Anything larger than 8x8 it will print the first 20 elements of the matrix.

The tiling matrix addition program will add two 2D matrices on the device GPU and CPU. The matrices that are calculated will be the same on the GPU as in the CPU. The GPU results will be transferred back to to the CPU. Both programs will compare the GPU-computed matrix with the CPU computed matrix and will print “Test PASSED” if they match.

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Matrix Add Screenshots:

```
[w859jas@gpu06 proj2]$ ./tileMatrixMult
Enter Matrix Size: 128
THREAD_BLOCK_SIZE: 8 x 8
MATRIX_SIZE: 128 x 128
GRID_SIZE: x: 16 y: 16
GPU time taken 0.087200 milliseconds
CPU time taken 2.760000 milliseconds
First 20 elements in GPU Matrix
54052240 53098760 53365078 54653504 52264906 52666713 55555334 53690342 56360209 53164194 50255966 56824424 51475758 55445607 57535826 54559503 53657362 54442269 53548014 52407322
First 20 elements in CPU Matrix
54052240 53098760 53365078 54653504 52264906 52666713 55555334 53690342 56360209 53164194 50255966 56824424 51475758 55445607 57535826 54559503 53657362 54442269 53548014 52407322
Test PASSED
```

```
[w859jas@gpu06 proj2]$ ./tileMatrixMult
Enter Matrix Size: 128
THREAD_BLOCK_SIZE: 16 x 16
MATRIX_SIZE: 128 x 128
GRID_SIZE: x: 8 y: 8
GPU time taken 0.086016 milliseconds
CPU time taken 2.749000 milliseconds
First 20 elements in GPU Matrix
54052240 53098760 53365078 54653504 52264906 52666713 55555334 53690342 56360209 53164194 50255966 56824424 51475758 55445607 57535826 54559503 53657362 54442269 53548014 52407322
First 20 elements in CPU Matrix
54052240 53098760 53365078 54653504 52264906 52666713 55555334 53690342 56360209 53164194 50255966 56824424 51475758 55445607 57535826 54559503 53657362 54442269 53548014 52407322
Test PASSED
```

```
Enter Matrix Size: 1024
THREAD_BLOCK_SIZE: 16 x 16
MATRIX_SIZE: 1024 x 1024
GRID_SIZE: x: 64 y: 64
GPU time taken 1.387840 milliseconds
CPU time taken 2970.524000 milliseconds
First 20 elements in GPU Matrix
414318491 394706972 4454357540 414468887 389263124 3893922947 426013124 383856155 407550238 433834835 424698901 4467223641 426737197 437891131 4377449091 3798754491 3770287921 3931616241 4165883811 3958183921
First 20 elements in CPU Matrix
414318491 394706972 4454357540 414468887 389263124 3893922947 426013124 383856155 407550238 433834835 424698901 4467223641 426737197 437891131 4377449091 3798754491 3770287921 3931616241 4165883811 3958183921
Test PASSED
[w859jas@gpu06 proj2]$ nvcc tileMatrixMult.cu -DTILE_WIDTH=8  -o tileMatrixMult
WARNING: group: unknown groupid 100000
[w859jas@gpu06 proj2]$ ./tileMatrixMult
Enter Matrix Size: 1024
THREAD_BLOCK_SIZE: 8 x 8
MATRIX_SIZE: 1024 x 1024
GRID_SIZE: x: 128 y: 128
GPU time taken 1.489312 milliseconds
CPU time taken 2894.949000 milliseconds
First 20 elements in GPU Matrix
414318491 394706972 4454357540 414468887 389263124 3893922947 426013124 383856155 407550238 433834835 424698901 4467223641 426737197 437891131 4377449091 3798754491 3770287921 3931616241 4165883811 3958183921
First 20 elements in CPU Matrix
414318491 394706972 4454357540 414468887 389263124 3893922947 426013124 383856155 407550238 433834835 424698901 4467223641 426737197 437891131 4377449091 3798754491 3770287921 3931616241 4165883811 3958183921
Test PASSED
```

```
[w859jas@gpu06 proj2]$ ./tileMatrixMult
Enter Matrix Size: 4096
THREAD_BLOCK_SIZE: 8 x 8
MATRIX_SIZE: 4096 x 4096
GRID_SIZE: x: 512 y: 512
GPU time taken 35.492702 milliseconds
CPU time taken 267034.313000 milliseconds
First 20 elements in GPU Matrix
1374327252 1268489876 2140736007 1947268169 1894421996 1866048005 1864104412 1211886924 1552957793 2134998102 1815855351 1711443999 1817100290 1715360588 1907351755 1400100037 1616687141 1307959165 1761708575 1893121393
First 20 elements in CPU Matrix
1374327252 1268489876 2140736007 1947268169 1894421996 1866048005 1864104412 1211886924 1552957793 2134998102 1815855351 1711443999 1817100290 1715360588 1907351755 1400100037 1616687141 1307959165 1761708575 1893121393
Test PASSED
```

```
[w859jas@gpu06 proj2]$ ./tileMatrixMult
Enter Matrix Size: 4096
THREAD_BLOCK_SIZE: 16 x 16
MATRIX_SIZE: 4096 x 4096
GRID_SIZE: x: 256 y: 256
GPU time taken 28.762432 milliseconds
CPU time taken 267392.111000 milliseconds
First 20 elements in GPU Matrix
1374327252 1268489876 2140736007 1947268169 1894421996 1866048005 1864104412 1211886924 1552957793 2134998102 1815855351 1711443999 1817100290 1715360588 1907351755 1400100037 1616687141 1307959165 1761708575 1893121393
First 20 elements in CPU Matrix
1374327252 1268489876 2140736007 1947268169 1894421996 1866048005 1864104412 1211886924 1552957793 2134998102 1815855351 1711443999 1817100290 1715360588 1907351755 1400100037 1616687141 1307959165 1761708575 1893121393
Test PASSED
```

```
[w859jas@gpu06 proj2]$ ./tileMatrixMult
Enter Matrix Size: 4096
THREAD_BLOCK_SIZE: 32 x 32
MATRIX_SIZE: 4096 x 4096
GRID_SIZE: x: 128 y: 128
GPU time taken 29.593439 milliseconds
CPU time taken 266901.022000 milliseconds
First 20 elements in GPU Matrix
1374327252 1268489876 2140736007 1947268169 1894421996 1866048005 1864104412 1211886924 1552957793 2134998102 1815855351 1711443999 1817100290 1715360588 1907351755 1400100037 1616687141 1307959165 1761708575 1893121393
First 20 elements in CPU Matrix
1374327252 1268489876 2140736007 1947268169 1894421996 1866048005 1864104412 1211886924 1552957793 2134998102 1815855351 1711443999 1817100290 1715360588 1907351755 1400100037 1616687141 1307959165 1761708575 1893121393
Test PASSED
```